

QF112 Regression Project

Jeffrey Gao

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Introduction of the Project

The goal of this project is to build a multiple linear regression model that predicts the monthly revenue of restaurants. The data file used is `restaurant_student_data.csv`, and the response variable is `monthly_revenue`. Extra variables I created were made directly from those variables, such as squared terms and interaction terms.

Setting the Data and Train/Testing It

First, I loaded the restaurant data into R. Then I made an 80% training set and a 20% testing set. The training set is used to build the model, while the testing set is used to check how well the model predicts new data. I used `set.seed(123)` so that the split stays the same every time the file is run.

```
restaurant_data <- read.csv("restaurant_student_data.csv")

set.seed(123)

train_practice <- sample(1:nrow(restaurant_data), size = 0.8 * nrow(restaurant_data))
traindata <- restaurant_data[train_practice, ]
testdata <- restaurant_data[-train_practice, ]

nrow(traindata)
```

```
## [1] 240
```

```
nrow(testdata)
```

```
## [1] 60
```

Data Summary

Before building the model, I looked at the structure and summary of the dataset. This helps show the response variable and the predictors available for the model.

```
str(restaurant_data)
```

```
## 'data.frame': 300 obs. of 20 variables:
## $ restaurant_id : int 1411 1087 3496 3323 333 1067 1307 2529 4710 3288 ...
## $ location_type : chr "downtown" "suburban" "downtown" "suburban" ...
## $ cuisine_type : chr "asian" "asian" "american" "asian" ...
## $ delivery : chr "yes" "yes" "no" "no" ...
## $ average_daily_walk_ins : int 34 55 113 32 35 86 50 31 99 84 ...
## $ monthly_marketing_spend: int 1277 1951 3208 500 1107 500 1215 500 2238 2620 ...
## $ seating_capacity : int 91 80 72 43 43 62 66 68 83 75 ...
## $ avg_check : num 72.7 29.9 63.5 37.1 57.6 ...
## $ star_rating : num 3 2.5 3.5 3 3.5 3 3 2.5 4 3.5 ...
## $ years_open : num 4.1 6.7 2 6.5 11.3 0.5 12.2 0.5 7.6 6.5 ...
## $ competitors_nearby : int 3 5 7 4 5 4 6 3 5 6 ...
## $ website_visits : int 484 629 1086 463 534 609 473 337 900 877 ...
## $ weekly_reservations : int 55 65 134 60 61 67 56 31 144 112 ...
## $ labor_hours : int 1478 1479 1670 1267 1113 1650 1327 1348 1713 1632 ...
## $ number_of_tables : int 40 35 39 24 28 33 31 28 45 39 ...
## $ table_turnover : num 0.8 0.99 1.15 1.08 1.1 0.87 0.8 0.89 1.11 1.31 ...
## $ food_cost_percent : num 0.276 0.292 0.304 0.259 0.251 0.272 0.301 0.299 0.325 0.321 ...
## $ avg_ticket_time : num 15.6 15.5 22.2 16.7 18.1 15.9 15.6 17 17.7 13.4 ...
## $ catering_inquiries : int 17 9 22 9 11 20 20 12 24 17 ...
## $ monthly_revenue : int 115196 104915 119446 80854 97077 111728 122198 84609 118679 105968
```

```
summary(restaurant_data)
```

```
## restaurant_id location_type cuisine_type delivery
## Min. : 28 Length:300 Length:300 Length:300
## 1st Qu.:1352 Class :character Class :character Class :character
## Median :2450 Mode :character Mode :character Mode :character
## Mean :2505
## 3rd Qu.:3611
## Max. :4989
## average_daily_walk_ins monthly_marketing_spend seating_capacity
## Min. : 15.00 Min. : 500 Min. : 20.00
## 1st Qu.: 56.00 1st Qu.:1627 1st Qu.: 72.00
## Median : 83.00 Median :2440 Median : 89.50
## Mean : 83.15 Mean :2475 Mean : 89.55
## 3rd Qu.:107.00 3rd Qu.:3138 3rd Qu.:106.25
## Max. :200.00 Max. :6116 Max. :159.00
## avg_check star_rating years_open competitors_nearby
## Min. : 18.00 Min. :2.500 Min. : 0.500 Min. : 0.000
## 1st Qu.: 35.73 1st Qu.:3.000 1st Qu.: 3.775 1st Qu.: 4.000
## Median : 57.69 Median :3.500 Median : 6.600 Median : 5.000
## Mean : 72.45 Mean :3.503 Mean : 6.910 Mean : 5.633
## 3rd Qu.: 90.35 3rd Qu.:4.000 3rd Qu.: 9.925 3rd Qu.: 8.000
## Max. :626.06 Max. :5.000 Max. :21.300 Max. :14.000
## website_visits weekly_reservations labor_hours number_of_tables
## Min. : 261.0 Min. : 30.0 Min. : 912 Min. :16.00
## 1st Qu.: 639.0 1st Qu.: 79.0 1st Qu.:1466 1st Qu.:36.00
## Median : 808.0 Median :110.5 Median :1632 Median :42.00
## Mean : 823.3 Mean :112.6 Mean :1635 Mean :41.63
## 3rd Qu.: 986.2 3rd Qu.:141.0 3rd Qu.:1796 3rd Qu.:48.00
## Max. :1569.0 Max. :227.0 Max. :2502 Max. :71.00
## table_turnover food_cost_percent avg_ticket_time catering_inquiries
## Min. :0.80 Min. :0.185 Min. :10.20 Min. : 3.00
```

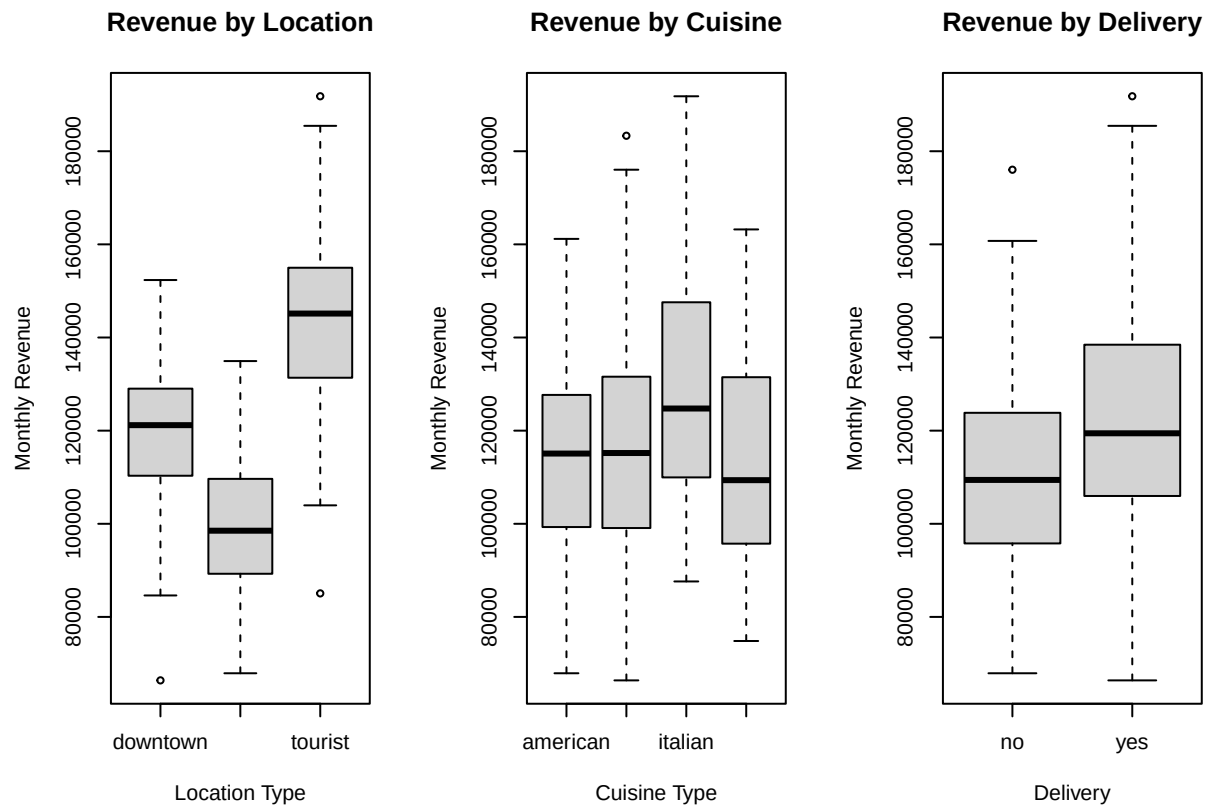
```
## 1st Qu.:0.93    1st Qu.:0.262    1st Qu.:16.30    1st Qu.:13.00
## Median :1.18    Median :0.291    Median :18.55    Median :17.50
## Mean   :1.23    Mean   :0.286    Mean   :18.77    Mean   :17.64
## 3rd Qu.:1.47    3rd Qu.:0.309    3rd Qu.:20.82    3rd Qu.:22.00
## Max.   :2.66    Max.   :0.389    Max.   :40.40    Max.   :40.00
## monthly_revenue
## Min.    : 66375
## 1st Qu.: 99759
## Median :116338
## Mean    :118606
## 3rd Qu.:133834
## Max.    :191800
```

The dataset has restaurant variables such as location type, cuisine type, delivery, average daily walk-ins, seating capacity, average check size, competitors nearby, weekly reservations, website visits, and monthly revenue. Since `monthly_revenue` is the variable being predicted, it is the dependent variable in the regression.

Boxplots

I used boxplots to compare monthly revenue across the categorical variables. This is useful because categorical variables can change the average level of revenue.

```
par(mfrow = c(1,3))
boxplot(monthly_revenue ~ location_type, data = restaurant_data,
        main = "Revenue by Location", xlab = "Location Type", ylab = "Monthly Revenue")
boxplot(monthly_revenue ~ cuisine_type, data = restaurant_data,
        main = "Revenue by Cuisine", xlab = "Cuisine Type", ylab = "Monthly Revenue")
boxplot(monthly_revenue ~ delivery, data = restaurant_data,
        main = "Revenue by Delivery", xlab = "Delivery", ylab = "Monthly Revenue")
```



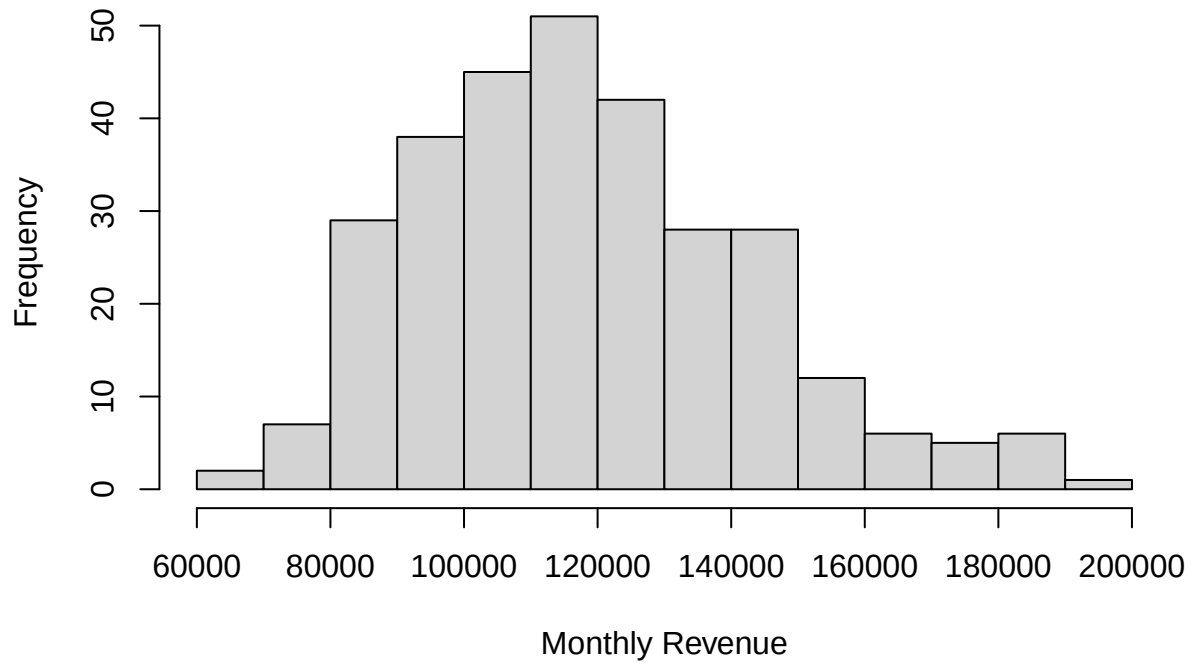
The boxplots show that monthly revenue changes depending on location type, cuisine type, and whether the restaurant offers delivery. Because of this, I kept these categorical variables in the model.

Data Analysis

Next, I looked at the distribution of monthly revenue and the relationships between monthly revenue and some important numeric predictors.

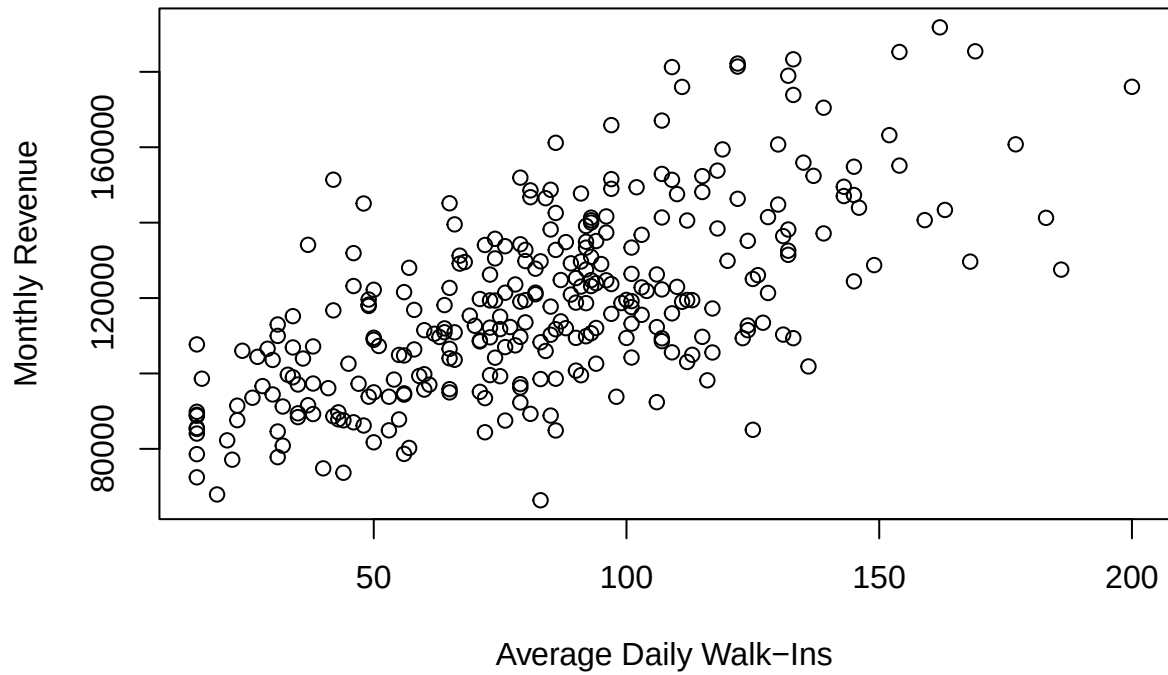
```
hist(restaurant_data$monthly_revenue,
     main = "Distribution of Monthly Revenue",
     xlab = "Monthly Revenue")
```

Distribution of Monthly Revenue



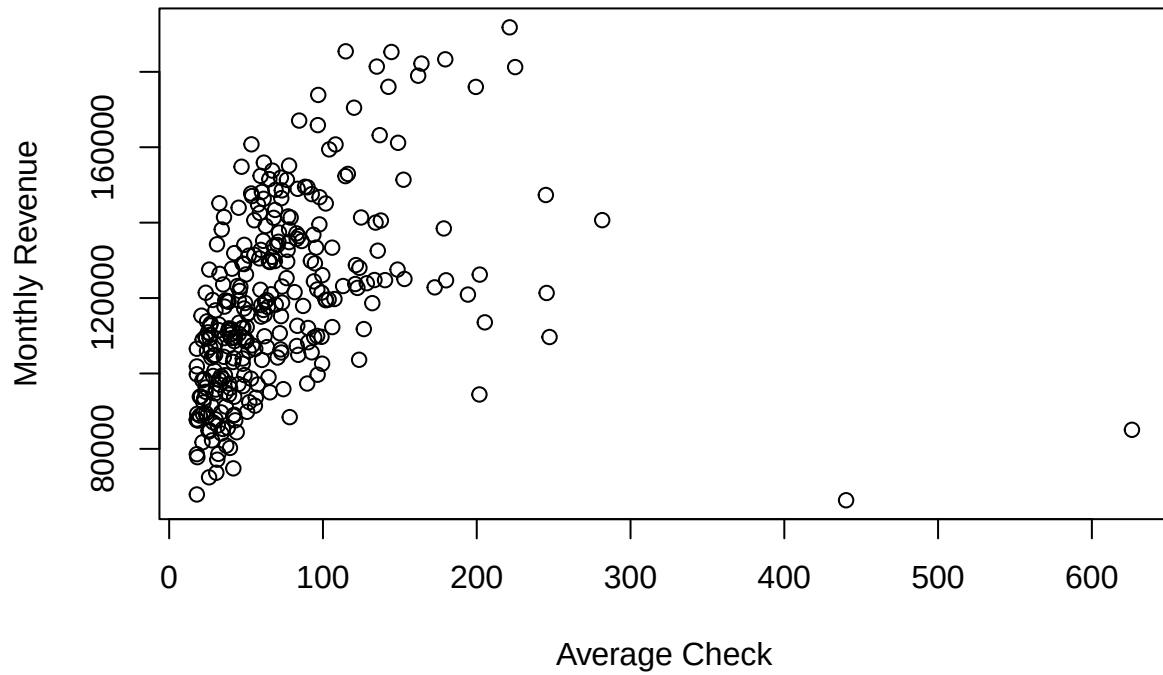
```
plot(restaurant_data$average_daily_walk_ins, restaurant_data$monthly_revenue,  
     main = "Revenue vs Average Daily Walk-Ins",  
     xlab = "Average Daily Walk-Ins", ylab = "Monthly Revenue")
```

Revenue vs Average Daily Walk-Ins

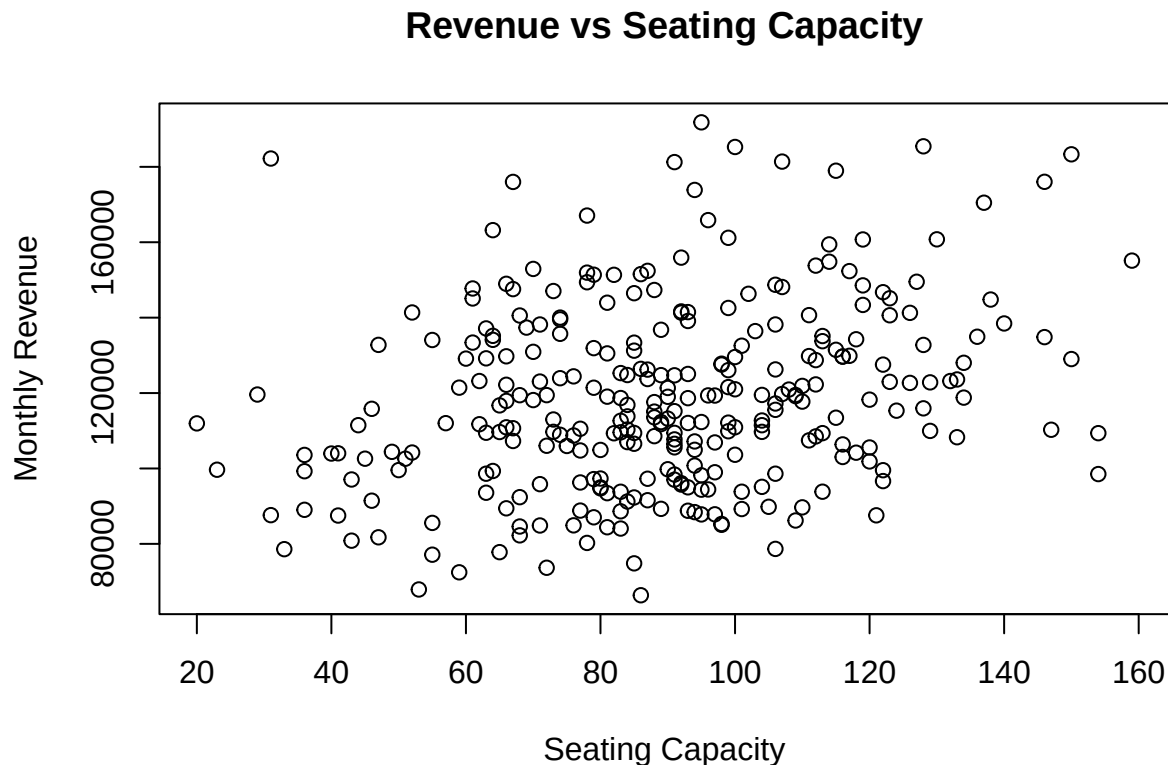


```
plot(restaurant_data$avg_check, restaurant_data$monthly_revenue,  
     main = "Revenue vs Average Check",  
     xlab = "Average Check", ylab = "Monthly Revenue")
```

Revenue vs Average Check



```
plot(restaurant_data$seating_capacity, restaurant_data$monthly_revenue,  
     main = "Revenue vs Seating Capacity",  
     xlab = "Seating Capacity", ylab = "Monthly Revenue")
```



Average daily walk-ins, average check, and seating capacity seem important because they are directly connected to how much money a restaurant can make. I also considered squared terms for average daily walk-ins and average check because the relationship did not look perfectly linear.

Multicollinearity Check

I checked correlations between the numeric predictors. Very high correlation between predictors can cause multicollinearity, which means some variables may be repeating the same information.

```
numeric_vars <- restaurant_data[, c("average_daily_walk_ins", "monthly_marketing_spend",
                                     "seating_capacity", "avg_check", "star_rating",
                                     "years_open", "competitors_nearby", "website_visits",
                                     "weekly_reservations", "labor_hours", "number_of_tables",
                                     "table_turnover", "food_cost_percent", "avg_ticket_time",
                                     "catering_inquiries")]

correlation_matrix <- cor(numeric_vars)
print(correlation_matrix)
```

```
##               average_daily_walk_ins  monthly_marketing_spend
## average_daily_walk_ins              1.0000000000             0.77083056
## monthly_marketing_spend             0.7708305561             1.00000000
## seating_capacity                    0.3785889022             0.44596432
## avg_check                          0.3213458124             0.30355724
```

## star_rating	0.4177316119	0.35516331		
## years_open	0.1452874898	0.19330928		
## competitors_nearby	0.7091353885	0.61940651		
## website_visits	0.9211029263	0.93784927		
## weekly_reservations	0.9281906842	0.73324671		
## labor_hours	0.7612407724	0.68267178		
## number_of_tables	0.5027188708	0.51258514		
## table_turnover	0.6518632436	0.56198571		
## food_cost_percent	0.0009748007	-0.03161414		
## avg_ticket_time	0.5361411305	0.46773353		
## catering_inquiries	0.7000466914	0.55829825		
##	seating_capacity	avg_check	star_rating	years_open
## average_daily_walk_ins	0.37858890	0.321345812	0.4177316	0.145287490
## monthly_marketing_spend	0.44596432	0.303557236	0.3551633	0.193309282
## seating_capacity	1.00000000	0.084538477	0.1622514	0.446241025
## avg_check	0.08453848	1.000000000	0.2068629	-0.004669346
## star_rating	0.16225139	0.206862871	1.0000000	0.149575405
## years_open	0.44624102	-0.004669346	0.1495754	1.000000000
## competitors_nearby	0.26424962	0.239257046	0.3495480	0.111774032
## website_visits	0.41546161	0.338227913	0.4539784	0.192386539
## weekly_reservations	0.34555721	0.336091201	0.5769476	0.143244315
## labor_hours	0.79201926	0.292547734	0.3680618	0.347548825
## number_of_tables	0.94378896	0.117095979	0.2174902	0.419895733
## table_turnover	0.09704689	0.285682259	0.6405762	0.072237982
## food_cost_percent	0.01378100	0.146983184	0.1626618	0.062670742
## avg_ticket_time	0.03582774	0.643178695	0.2519106	-0.105500573
## catering_inquiries	0.23953846	0.581675251	0.3050306	0.073110116
##	competitors_nearby	website_visits	weekly_reservations	
## average_daily_walk_ins	0.70913539	0.921102926	0.92819068	
## monthly_marketing_spend	0.61940651	0.937849266	0.73324671	
## seating_capacity	0.26424962	0.415461608	0.34555721	
## avg_check	0.23925705	0.338227913	0.33609120	
## star_rating	0.34954804	0.453978364	0.57694757	
## years_open	0.11177403	0.192386539	0.14324431	
## competitors_nearby	1.00000000	0.699511118	0.68280178	
## website_visits	0.69951112	1.000000000	0.87740155	
## weekly_reservations	0.68280178	0.877401548	1.00000000	
## labor_hours	0.53893881	0.751240389	0.71408841	
## number_of_tables	0.36765958	0.515535324	0.45784779	
## table_turnover	0.53812241	0.660745177	0.71652373	
## food_cost_percent	0.03213094	-0.009450314	0.02911174	
## avg_ticket_time	0.44795631	0.525537087	0.51350078	
## catering_inquiries	0.49407137	0.657819618	0.66529224	
##	labor_hours	number_of_tables	table_turnover	
## average_daily_walk_ins	0.76124077	0.50271887	0.65186324	
## monthly_marketing_spend	0.68267178	0.51258514	0.56198571	
## seating_capacity	0.79201926	0.94378896	0.09704689	
## avg_check	0.29254773	0.11709598	0.28568226	
## star_rating	0.36806178	0.21749016	0.64057618	
## years_open	0.34754882	0.41989573	0.07223798	
## competitors_nearby	0.53893881	0.36765958	0.53812241	
## website_visits	0.75124039	0.51553532	0.66074518	
## weekly_reservations	0.71408841	0.45784779	0.71652373	
## labor_hours	1.00000000	0.82592652	0.43224865	

```
## number_of_tables      0.82592652      1.00000000      0.18933991
## table_turnover        0.43224865      0.18933991      1.00000000
## food_cost_percent     0.01266535      0.05138734      0.13681779
## avg_ticket_time       0.32574209      0.10887865      0.45851985
## catering_inquiries    0.53591650      0.32425977      0.47710545
##
## food_cost_percent     0.0009748007      0.53614113      0.70004669
## average_daily_walk_ins 0.0009748007      0.53614113      0.70004669
## monthly_marketing_spend -0.0316141416      0.46773353      0.55829825
## seating_capacity      0.0137809950      0.03582774      0.23953846
## avg_check             0.1469831838      0.64317869      0.58167525
## star_rating           0.1626617651      0.25191065      0.30503060
## years_open            0.0626707416     -0.10550057      0.07311012
## competitors_nearby    0.0321309383      0.44795631      0.49407137
## website_visits        -0.0094503143      0.52553709      0.65781962
## weekly_reservations   0.0291117375      0.51350078      0.66529224
## labor_hours           0.0126653533      0.32574209      0.53591650
## number_of_tables      0.0513873391      0.10887865      0.32425977
## table_turnover        0.1368177901      0.45851985      0.47710545
## food_cost_percent     1.0000000000     -0.01649239      0.02919171
## avg_ticket_time       -0.0164923900      1.00000000      0.55846378
## catering_inquiries    0.0291917147      0.55846378      1.00000000
```

```
highcorrelation <- which(abs(correlation_matrix) > 0.8 & abs(correlation_matrix) < 1,
                          arr.ind = TRUE)
```

```
if (nrow(highcorrelation) > 0) {
  cat("Potential multicollinearity was found:\n")
  apply(highcorrelation, 1, function(row) {
    cat(rownames(correlation_matrix)[row[1]], "and",
        colnames(correlation_matrix)[row[2]],
        "with correlation", correlation_matrix[row[1], row[2]], "\n")
  })
} else {
  cat("No serious multicollinearity was found.\n")
}
```

```
## Potential multicollinearity was found:
## website_visits and average_daily_walk_ins with correlation 0.9211029
## weekly_reservations and average_daily_walk_ins with correlation 0.9281907
## website_visits and monthly_marketing_spend with correlation 0.9378493
## number_of_tables and seating_capacity with correlation 0.943789
## average_daily_walk_ins and website_visits with correlation 0.9211029
## monthly_marketing_spend and website_visits with correlation 0.9378493
## weekly_reservations and website_visits with correlation 0.8774015
## average_daily_walk_ins and weekly_reservations with correlation 0.9281907
## website_visits and weekly_reservations with correlation 0.8774015
## number_of_tables and labor_hours with correlation 0.8259265
## seating_capacity and number_of_tables with correlation 0.943789
## labor_hours and number_of_tables with correlation 0.8259265

## NULL
```

This step was used to understand whether the numeric variables were strongly related to each other. I did

not remove variables only because of correlation, but I used this information when deciding how complicated the final model should be.

Model 1: Basic Model

I first made a basic multiple linear regression model using most of the available predictors. This gave me a starting point to compare against later models.

```
modell1 <- lm(
  monthly_revenue ~ location_type + cuisine_type + delivery +
    average_daily_walk_ins + monthly_marketing_spend + seating_capacity +
    avg_check + star_rating + years_open + competitors_nearby +
    website_visits + weekly_reservations + labor_hours + number_of_tables +
    table_turnover + food_cost_percent + avg_ticket_time + catering_inquiries,
  data = traindata
)

summary(modell1)
```

```
##
## Call:
## lm(formula = monthly_revenue ~ location_type + cuisine_type +
##     delivery + average_daily_walk_ins + monthly_marketing_spend +
##     seating_capacity + avg_check + star_rating + years_open +
##     competitors_nearby + website_visits + weekly_reservations +
##     labor_hours + number_of_tables + table_turnover + food_cost_percent +
##     avg_ticket_time + catering_inquiries, data = traindata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -58173  -5199    -72    4991   25079
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      6.513e+04  9.906e+03   6.574 3.55e-10 ***
## location_typesuburban -1.772e+04  1.470e+03 -12.059 < 2e-16 ***
## location_typetourist  1.879e+04  1.598e+03  11.755 < 2e-16 ***
## cuisine_typeasian    5.036e+03  1.596e+03   3.155 0.00183 **
## cuisine_typeitalian  1.238e+04  1.683e+03   7.357 3.77e-12 ***
## cuisine_typemexican  -3.418e+03  1.929e+03  -1.772 0.07775 .
## deliveryyes         1.072e+04  1.201e+03   8.923 < 2e-16 ***
## average_daily_walk_ins 1.746e+02  7.231e+01   2.415 0.01655 *
## monthly_marketing_spend -5.788e-01  2.143e+00  -0.270 0.78733
## seating_capacity      1.801e+01  8.290e+01   0.217 0.82821
## avg_check           7.617e+00  1.587e+01   0.480 0.63169
## star_rating          2.064e+03  1.532e+03   1.347 0.17928
## years_open          -3.620e+02  1.430e+02  -2.532 0.01204 *
## competitors_nearby    -1.368e+03  2.854e+02  -4.794 3.03e-06 ***
## website_visits        1.420e+01  1.531e+01   0.928 0.35469
## weekly_reservations    9.267e+01  4.379e+01   2.116 0.03547 *
## labor_hours           5.149e+00  6.133e+00   0.840 0.40209
## number_of_tables      1.328e+02  2.045e+02   0.650 0.51654
```

```
## table_turnover      -3.098e+03  2.686e+03  -1.153  0.25004
## food_cost_percent   1.001e+04  2.215e+04   0.452  0.65168
## avg_ticket_time     6.101e+00  2.359e+02   0.026  0.97939
## catering_inquiries  -1.171e+02  1.432e+02  -0.818  0.41448
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8326 on 218 degrees of freedom
## Multiple R-squared:  0.8929, Adjusted R-squared:  0.8825
## F-statistic: 86.5 on 21 and 218 DF,  p-value: < 2.2e-16
```

Model 2: Model with Transformations

The second model adds squared terms for average daily walk-ins and average check. These transformations are allowed because they are made directly from variables in the dataset.

```
traindata$walk_ins_2 <- traindata$average_daily_walk_ins^2
testdata$walk_ins_2 <- testdata$average_daily_walk_ins^2

traindata$avg_check_2 <- traindata$avg_check^2
testdata$avg_check_2 <- testdata$avg_check^2

model2 <- lm(
  monthly_revenue ~ location_type + cuisine_type + delivery +
    average_daily_walk_ins + walk_ins_2 + seating_capacity +
    avg_check + avg_check_2 + competitors_nearby + years_open +
    weekly_reservations + website_visits + food_cost_percent,
  data = traindata
)

summary(model2)
```

```
##
## Call:
## lm(formula = monthly_revenue ~ location_type + cuisine_type +
##     delivery + average_daily_walk_ins + walk_ins_2 + seating_capacity +
##     avg_check + avg_check_2 + competitors_nearby + years_open +
##     weekly_reservations + website_visits + food_cost_percent,
##     data = traindata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16199.1  -4500.7  -244.6   3921.9  19072.1
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    5.908e+04  5.850e+03  10.099 < 2e-16 ***
## location_typesuburban -1.285e+04  1.148e+03 -11.195 < 2e-16 ***
## location_typedtourist  1.976e+04  1.212e+03  16.298 < 2e-16 ***
## cuisine_typeasian     4.627e+03  1.250e+03   3.702 0.000269 ***
## cuisine_typeitalian    8.348e+03  1.355e+03   6.159 3.37e-09 ***
## cuisine_typemexican   -2.355e+03  1.488e+03  -1.582 0.114964
```

```
## deliveryyes          1.170e+04  9.423e+02  12.414 < 2e-16 ***
## average_daily_walk_ins 1.817e+02  5.804e+01  3.131 0.001974 **
## walk_ins_2            5.967e-02  2.331e-01  0.256 0.798191
## seating_capacity      8.918e+01  2.034e+01  4.385 1.79e-05 ***
## avg_check             2.681e+02  2.432e+01  11.024 < 2e-16 ***
## avg_check_2           -8.238e-01  7.019e-02 -11.736 < 2e-16 ***
## competitors_nearby    -1.169e+03  2.212e+02  -5.286 2.98e-07 ***
## years_open            -2.198e+02  1.108e+02  -1.984 0.048537 *
## weekly_reservations    4.119e+01  2.981e+01  1.382 0.168507
## website_visits         1.176e+01  4.438e+00  2.649 0.008639 **
## food_cost_percent      2.496e+04  1.717e+04  1.454 0.147369
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6549 on 223 degrees of freedom
## Multiple R-squared:  0.9322, Adjusted R-squared:  0.9273
## F-statistic: 191.6 on 16 and 223 DF,  p-value: < 2.2e-16
```

Model 3: Final Interaction Model

The final model includes variables that make business sense and adds interaction terms. For example, the effect of average daily walk-ins may depend on location type, and the effect of average check may depend on cuisine type or location type. Website visits and weekly reservations may also depend on whether delivery is offered.

```
model3 <- lm(
  monthly_revenue ~ location_type + cuisine_type + delivery +
    average_daily_walk_ins + walk_ins_2 + seating_capacity +
    avg_check + avg_check_2 + competitors_nearby + years_open +
    weekly_reservations + website_visits + food_cost_percent +
    average_daily_walk_ins:location_type +
    avg_check:cuisine_type +
    avg_check:location_type +
    weekly_reservations:delivery +
    website_visits:delivery +
    avg_check:food_cost_percent,
  data = traindata
)

summary(model3)

##
## Call:
## lm(formula = monthly_revenue ~ location_type + cuisine_type +
##     delivery + average_daily_walk_ins + walk_ins_2 + seating_capacity +
##     avg_check + avg_check_2 + competitors_nearby + years_open +
##     weekly_reservations + website_visits + food_cost_percent +
##     average_daily_walk_ins:location_type + avg_check:cuisine_type +
##     avg_check:location_type + weekly_reservations:delivery +
##     website_visits:delivery + avg_check:food_cost_percent, data = traindata)
##
## Residuals:
```

```

##      Min      1Q   Median      3Q      Max
## -13801.3 -3453.2    -2.3   3034.4 12562.4
##
## Coefficients:
##
##              Estimate Std. Error t value
## (Intercept)      8.416e+04  7.559e+03  11.134
## location_typesuburban -1.444e+04  2.464e+03  -5.863
## location_typedtourist  3.749e+03  2.714e+03   1.381
## cuisine_typeasian     4.336e+03  1.802e+03   2.406
## cuisine_typeitalian    1.027e+04  2.054e+03   5.000
## cuisine_typedmexican  -5.575e+03  1.926e+03  -2.895
## deliveryyes          3.653e+03  2.295e+03   1.591
## average_daily_walk_ins  2.580e+02  5.205e+01   4.957
## walk_ins_2          -3.543e-01  2.083e-01  -1.701
## seating_capacity      1.069e+02  1.646e+01   6.497
## avg_check           -1.226e+01  9.022e+01  -0.136
## avg_check_2          -6.948e-01  8.539e-02  -8.138
## competitors_nearby    -1.099e+03  1.809e+02  -6.077
## years_open           -2.924e+02  9.044e+01  -3.233
## weekly_reservations    1.377e+01  3.668e+01   0.375
## website_visits        8.044e+00  5.291e+00   1.520
## food_cost_percent     -3.292e+04  2.424e+04  -1.358
## location_typesuburban:average_daily_walk_ins -2.545e+01  2.644e+01  -0.963
## location_typedtourist:average_daily_walk_ins  8.581e+00  2.845e+01   0.302
## cuisine_typeasian:avg_check -2.919e-01  2.235e+01  -0.013
## cuisine_typeitalian:avg_check -3.970e+01  2.290e+01  -1.734
## cuisine_typedmexican:avg_check  4.507e+01  2.626e+01   1.716
## location_typesuburban:avg_check  2.053e+01  2.398e+01   0.856
## location_typedtourist:avg_check  1.878e+02  2.349e+01   7.995
## deliveryyes:weekly_reservations  4.613e+01  3.863e+01   1.194
## deliveryyes:website_visits    2.961e+00  5.822e+00   0.509
## avg_check:food_cost_percent  6.661e+02  3.169e+02   2.102
##
##              Pr(>|t|)
## (Intercept)      < 2e-16 ***
## location_typesuburban 1.71e-08 ***
## location_typedtourist  0.16867
## cuisine_typeasian     0.01697 *
## cuisine_typeitalian    1.20e-06 ***
## cuisine_typedmexican    0.00419 **
## deliveryyes          0.11300
## average_daily_walk_ins 1.46e-06 ***
## walk_ins_2          0.09046 .
## seating_capacity      5.71e-10 ***
## avg_check            0.89205
## avg_check_2          3.33e-14 ***
## competitors_nearby    5.58e-09 ***
## years_open           0.00142 **
## weekly_reservations    0.70776
## website_visits        0.12988
## food_cost_percent     0.17579
## location_typesuburban:average_daily_walk_ins  0.33679
## location_typedtourist:average_daily_walk_ins  0.76324
## cuisine_typeasian:avg_check  0.98959
## cuisine_typeitalian:avg_check  0.08445 .

```

```
## cuisine_typemexican:avg_check          0.08756 .
## location_typesuburban:avg_check        0.39298
## location_typetourist:avg_check         8.12e-14 ***
## deliveryyes:weekly_reservations        0.23384
## deliveryyes:website_visits             0.61153
## avg_check:food_cost_percent            0.03676 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5206 on 213 degrees of freedom
## Multiple R-squared:  0.9591, Adjusted R-squared:  0.9541
## F-statistic: 191.9 on 26 and 213 DF,  p-value: < 2.2e-16
```

RMSE and MSE

I used MSE and RMSE to compare the models. The project says the final model will be evaluated using mean squared error, so MSE is the most important comparison. RMSE is also useful because it is in the same units as monthly revenue.

```
MSE <- function(actual, predicted) {
  mean((actual - predicted)^2)
}

RMSE <- function(actual, predicted) {
  sqrt(mean((actual - predicted)^2))
}

pred1 <- predict(model1, newdata = testdata)
pred2 <- predict(model2, newdata = testdata)
pred3 <- predict(model3, newdata = testdata)

mse1 <- MSE(testdata$monthly_revenue, pred1)
mse2 <- MSE(testdata$monthly_revenue, pred2)
mse3 <- MSE(testdata$monthly_revenue, pred3)

rmse1 <- RMSE(testdata$monthly_revenue, pred1)
rmse2 <- RMSE(testdata$monthly_revenue, pred2)
rmse3 <- RMSE(testdata$monthly_revenue, pred3)

model_comparison <- data.frame(
  Model = c("Basic Model", "Transformation Model", "Final Interaction Model"),
  Test_MSE = c(mse1, mse2, mse3),
  Test_RMSE = c(rmse1, rmse2, rmse3)
)

model_comparison
```

	Model	Test_MSE	Test_RMSE
## 1	Basic Model	143958846	11998.29
## 2	Transformation Model	202488933	14229.86
## 3	Final Interaction Model	30355032	5509.54

The final interaction model was chosen because it uses important restaurant variables, includes reasonable transformations, and allows important relationships to change depending on restaurant type. It is still a multiple linear regression model and uses only `lm`.

Final Model on the Full Dataset

After choosing the final model, I fit the same model on the full dataset. This is the model structure used in the submitted `predict_revenue` function.

```
restaurant_data$walk_ins_2 <- restaurant_data$average_daily_walk_ins^2
restaurant_data$avg_check_2 <- restaurant_data$avg_check^2

finalmodel <- lm(
  monthly_revenue ~ location_type + cuisine_type + delivery +
    average_daily_walk_ins + walk_ins_2 + seating_capacity +
    avg_check + avg_check_2 + competitors_nearby + years_open +
    weekly_reservations + website_visits + food_cost_percent +
    average_daily_walk_ins:location_type +
    avg_check:cuisine_type +
    avg_check:location_type +
    weekly_reservations:delivery +
    website_visits:delivery +
    avg_check:food_cost_percent,
  data = restaurant_data
)

summary(finalmodel)
```

```
##
## Call:
## lm(formula = monthly_revenue ~ location_type + cuisine_type +
##     delivery + average_daily_walk_ins + walk_ins_2 + seating_capacity +
##     avg_check + avg_check_2 + competitors_nearby + years_open +
##     weekly_reservations + website_visits + food_cost_percent +
##     average_daily_walk_ins:location_type + avg_check:cuisine_type +
##     avg_check:location_type + weekly_reservations:delivery +
##     website_visits:delivery + avg_check:food_cost_percent, data = restaurant_data)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-14605.9	-3427.0	-72.2	3010.7	12954.8

```
##
## Coefficients:
```

	Estimate	Std. Error	t value
(Intercept)	7.781e+04	6.667e+03	11.670
location_typesuburban	-1.413e+04	2.165e+03	-6.529
location_typedtourist	5.596e+03	2.322e+03	2.410
cuisine_typeasian	5.622e+03	1.520e+03	3.699
cuisine_typeitalian	1.027e+04	1.816e+03	5.656
cuisine_typemexican	-4.074e+03	1.718e+03	-2.371
deliveryyes	3.287e+03	2.064e+03	1.592
average_daily_walk_ins	2.611e+02	4.661e+01	5.602

```

## walk_ins_2 -3.576e-01 1.933e-01 -1.850
## seating_capacity 1.128e+02 1.433e+01 7.873
## avg_check 5.429e+01 7.480e+01 0.726
## avg_check_2 -6.574e-01 4.438e-02 -14.813
## competitors_nearby -1.195e+03 1.594e+02 -7.496
## years_open -2.674e+02 8.213e+01 -3.256
## weekly_reservations 2.952e+01 3.108e+01 0.950
## website_visits 6.448e+00 4.597e+00 1.403
## food_cost_percent -1.533e+04 2.030e+04 -0.755
## location_typesuburban:average_daily_walk_ins -2.233e+01 2.427e+01 -0.920
## location_typedtourist:average_daily_walk_ins -2.239e+00 2.522e+01 -0.089
## cuisine_typeasian:avg_check -1.664e+01 1.921e+01 -0.866
## cuisine_typeitalian:avg_check -3.204e+01 1.923e+01 -1.666
## cuisine_typedmexican:avg_check 2.182e+01 2.266e+01 0.963
## location_typesuburban:avg_check 1.090e+01 1.864e+01 0.585
## location_typedtourist:avg_check 1.732e+02 1.423e+01 12.173
## deliveryyes:weekly_reservations 3.356e+01 3.314e+01 1.013
## deliveryyes:website_visits 4.768e+00 5.140e+00 0.928
## avg_check:food_cost_percent 4.451e+02 2.301e+02 1.935
## Pr(>|t|)
## (Intercept) < 2e-16 ***
## location_typesuburban 3.21e-10 ***
## location_typedtourist 0.016624 *
## cuisine_typeasian 0.000262 ***
## cuisine_typeitalian 3.91e-08 ***
## cuisine_typedmexican 0.018411 *
## deliveryyes 0.112520
## average_daily_walk_ins 5.17e-08 ***
## walk_ins_2 0.065341 .
## seating_capacity 8.22e-14 ***
## avg_check 0.468550
## avg_check_2 < 2e-16 ***
## competitors_nearby 9.24e-13 ***
## years_open 0.001271 **
## weekly_reservations 0.343088
## website_visits 0.161836
## food_cost_percent 0.450906
## location_typesuburban:average_daily_walk_ins 0.358505
## location_typedtourist:average_daily_walk_ins 0.929342
## cuisine_typeasian:avg_check 0.387125
## cuisine_typeitalian:avg_check 0.096788 .
## cuisine_typedmexican:avg_check 0.336391
## location_typesuburban:avg_check 0.559337
## location_typedtourist:avg_check < 2e-16 ***
## deliveryyes:weekly_reservations 0.312131
## deliveryyes:website_visits 0.354422
## avg_check:food_cost_percent 0.054051 .
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5213 on 273 degrees of freedom
## Multiple R-squared: 0.9583, Adjusted R-squared: 0.9544
## F-statistic: 241.4 on 26 and 273 DF, p-value: < 2.2e-16

```

Testing the Function File

The project also requires a separate .R file with a function named `predict_revenue`. I tested the function on the example new data file to make sure it returns predictions.

```
source("predict_revenue.R")
new_data_example <- read.csv("new_data_example.csv")

example_predictions <- predict_revenue(restaurant_data, new_data_example)
example_predictions
```

```
## [1] 100718.90 100478.77 89765.98 88337.51
```

```
is.numeric(example_predictions)
```

```
## [1] TRUE
```

```
length(example_predictions)
```

```
## [1] 4
```

The function returns a numeric vector, and the number of predictions matches the number of rows in the new data file. This means the function follows the required format and should work on new unseen data with the same predictor columns.

Summary

In conclusion, I built a multiple linear regression model to predict restaurant monthly revenue. I started with a train/test split, looked at summaries, boxplots, plots, and correlations, and then compared multiple models. The final model uses categorical variables, squared terms, and interaction terms while staying inside the `lm` framework. The final model includes location type and cuisine type because restaurant revenue may differ by market and restaurant category. It includes seating capacity, competitors nearby, and years open as general business characteristics. The squared terms allow curved relationships for walk-ins and average check. The interaction terms allow the impact of walk-ins, average check, reservations, website visits, and food cost percentage to depend on restaurant context.

Test the Submitted Function

The submitted .R file contains one function named `predict_revenue`. The function fits the final model on `old_data` and returns predictions for `new_data`.

```
source("predict_revenue.R")
new_data_example <- read.csv("new_data_example.csv")

example_predictions <- predict_revenue(restaurant_data, new_data_example)
example_predictions
```

```
## [1] 100718.90 100478.77 89765.98 88337.51
```

```
is.numeric(example_predictions)
```

```
## [1] TRUE
```

```
length(example_predictions)
```

```
## [1] 4
```

The function returns a numeric vector with one prediction for each row of the new dataset.